

Computer Network Defense: Firewall Lab

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1 Introduction

A **firewall** is a security system that monitors and controls incoming and outgoing network traffic based on a set of rules. Firewalls act as a barrier between a trusted internal network and untrusted external networks, stopping unauthorized access from outside while also preventing malware from communicating out.

In this lab you will configure a Linux firewall on your virtual machine using **UFW** (Uncomplicated Firewall). UFW is a command-line tool that simplifies the management of firewall rules on Linux systems. By the end of this lab your group will have set up a working firewall, allowed and denied specific types of traffic, and tested those rules across your VMs.

Note

All commands in this lab should be run on your **virtual machine**, not your host machine.

2 Installing UFW

First, make sure your system is up to date, then install UFW.

```
sudo apt update
sudo apt install ufw
```

Optionally, if you prefer a graphical interface, you can also install GFW. These instructions only use the command-line version above.

```
sudo apt install gufw
```

This lab tests the firewall using SSH, so your VM also needs to be running an SSH server. Ubuntu desktop does not include one by default, so install and start it now.

```
sudo apt install openssh-server
sudo systemctl enable --now ssh
```

3 Setting Up Default Rules

We will start by enabling the firewall and blocking all incoming traffic by default.

1. Enable the firewall.

```
sudo ufw enable
```

2. Check the firewall status.

```
sudo ufw status verbose
```

Expected Result

```
Status: active
Logging: on (low)
Default: deny (incoming), allow (outgoing), disable (routed)
New Profiles: skip
```

3. If incoming traffic isn't already denied, run the following command to deny all incoming connections.

```
sudo ufw default deny incoming
```

4. Run `sudo ufw status verbose` again to confirm incoming traffic is being denied.

! Testing SSH Across the Group

Confirm that SSH isn't allowed by having someone in your group try to ssh into your machine. You will need to share your ip address with them, so be sure to run the following command to get it.

```
ip a
```

Then have them try to connect to your machine from their machine

```
ssh <your_username>@<your_ip_address>
```

Confirm that they are blocked. Because UFW silently drops the blocked packets, the connection attempt should hang and eventually **time out**. If instead they immediately see **Connection refused**, the packets reached your machine but nothing was listening on port 22. Check that the SSH server from the installation step is running with `systemctl status ssh`.

! Group Discussion

With your firewall now blocking all incoming traffic, discuss the following with your group:

1. What does it mean practically for a machine to deny all incoming traffic? What can it still do?
2. Can your VM still access the internet (outgoing traffic)? Try it by opening a browser or run `ping google.com`.
3. What is a scenario where blocking all incoming traffic by default would cause a problem?

4 Allowing a Specific Service

Blocking everything is not practical. Let's allow SSH connections so that other machines can connect to yours.

1. Allow SSH traffic through your firewall.

```
sudo ufw allow ssh
```

2. Check your updated firewall status.

```
sudo ufw status verbose
```

! Testing SSH Again

Now that SSH is allowed, test it again with another group member. Share your ip address with a group member.

```
ip a
```

Then have them connect to your machine with the following command

```
ssh <your_username>@<your_ip_address>
```

They should be prompted for your password. Perhaps just type it in on their computer instead of telling them what it is. Once you've typed it in, they should be connected! To disconnect, type `exit`.

5 Removing a Firewall Rule

Firewalls need to be maintained. You may need to remove rules that are no longer needed.

Option 1: Remove a rule by service name.

```
sudo ufw delete allow ssh
```

Option 2: List all rules by number, then delete by number.

```
sudo ufw status numbered  
sudo ufw delete <rule_number>
```

Tip

Use Option 2 when you are unsure of the exact rule name, or when you have multiple rules and want to be precise about which one to remove.

6 Denying Traffic from a Specific Source

You can also block traffic from a specific IP address or subnet. This is useful for blocking a known bad actor.

```
sudo ufw insert 1 deny from <ip_address>          # deny from a single IP
sudo ufw insert 1 deny from <network_address>/24 # deny from an entire
↪ subnet, e.g. 192.168.1.0/24
```

Warning

UFW checks rules **in order** and stops at the first rule that matches. If you simply ran `sudo ufw deny from <ip_address>`, the new rule would be added to the end of the list, *after* the rule allowing SSH, so SSH traffic from that address would still be allowed. `insert 1` places the deny rule at the top of the list so that it is checked first. Use `sudo ufw status numbered` to see the order of your rules.

Group Activity

Work with another group member to try the following:

1. Find your partner's IP address (`ip a`).
2. On your machine, remove the general SSH allow rule and replace it with one that only allows SSH from your partner's IP address.

```
sudo ufw delete allow ssh
sudo ufw allow from <partner_ip_address> to any port 22 proto tcp
```

3. Have them attempt to SSH into your machine. Does it work?
4. Have a *different* group member (one not in your rule) attempt to SSH in. What is the result?

7 Cleanup and Turning off the Firewall

If you need to temporarily disable your firewall and suspend all active rules, run the following. Note that disabling the firewall keeps your rules, so they will come back the next time the firewall is enabled.

```
sudo ufw disable
```

To clean up the changes we made in this lab so they do not affect future labs in this course, reset UFW. This disables the firewall **and** deletes all of the rules you added. You will be asked to confirm.

```
sudo ufw reset
```

8 Wrap-Up Discussion

! Group Assignment

Before you leave, discuss the **Firewall Lab** questions in the *Computer Network Defense: Assignment* document with your group and write down your answers. Your group will submit the answers for all of the labs together, as a single PDF on Canvas, after the final lab.

9 Additional Resources

For more UFW commands and examples, see [Digital Ocean's UFW Essentials article](#).